Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

Division C | Astronomy | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) What can power a protostar before sustained fusion?

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

A. A planetary nebula necessarily

B. No energy source

C. Gravitational contraction

D. Chemical oxygen combustion

Correct: C

2. (2 points) Hotter blackbodies peak at…

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

A. The same wavelength always

B. Zero frequency

C. Shorter wavelength

D. Longer wavelength

Correct: C

3. (2 points) An isochrone describes…

Unit: Advanced graph and image interpretation.
Worked-case data: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

A. Equal age across modeled masses

B. One star’s spatial orbit

C. Only equal radii

D. A single observed pulse

Correct: A

4. (2 points) A black-hole horizon is…

Unit: Supernovae, white dwarfs, neutron stars, and black holes.
Worked-case data: Use 3 km per solar mass as a rounded Schwarzschild-radius scale for a 10 Msun nonrotating black hole.

A. A causal boundary

B. A normal rocky surface

C. A visible shell of gas necessarily

D. The edge of every accretion disk

Correct: A

5. (4 points) Compute the example luminosity.

Unit: Stellar properties, fusion, and the H-R diagram.
Worked-case data: A star has radius 2 Rsun and temperature 2 Tsun. Calculate L/Lsun.

answer: 2^2 x 2^4 = 64 Lsun. (4 points).

6. (4 points) Solve the example distance modulus.

Unit: Astrophysics equations and quantitative synthesis.
Worked-case data: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

answer: d = 10 x 10^((10-5)/5) = 100 pc. (4 points).

7. (4 points) Distinguish emission and reflection nebulae.

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

answer: Emission produces radiation from excited gas; reflection redirects radiation from another source. (4 points).

8. (4 points) What does planetary nebula not imply?

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

answer: It does not mean the nebula formed from planets; the term is historical. (4 points).

9. (4 points) What can a nondetection mean?

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

answer: The component may be absent, too faint, unresolved, obscured, or outside the observed wavelength sensitivity. (4 points).

10. (6 points) Solve this case and explain your reasoning: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

reasoning: At fixed T, L □ R^2.
R2/R1 = sqrt(100/1) = 10.
Equal apparent brightness would not support this conclusion without distance information. (3 points).

conclusion: The more luminous star has ten times the radius under the stated model. (3 points).

11. (2 points) A synthetic star has parallax 0.003333333 arcseconds. Use d(pc)=1/p to find distance.

answer: 300 pc (2 points).

Numerical target 300; absolute tolerance 0.6. Accepted units: , pc.

12. (2 points) A standard-candle exercise gives apparent magnitude 13, absolute magnitude 3 and negligible extinction. Use m-M=5log10(d/10 pc). Find d.

answer: 1000 pc (2 points).

Numerical target 1000; absolute tolerance 2. Accepted units: , pc.

13. (2 points) A star has twice the reference radius and the same temperature. Find its luminosity ratio using L proportional to R^2 T^4.

answer: 4 (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , .

14. (2 points) Use Wien's constant 2.90e-3 m K for a 5,800 K star. Find peak wavelength.

answer: 5e-7 m (2 points).

Numerical target 5e-07; absolute tolerance 1e-09. Accepted units: , m.

15. (2 points) An artificial galaxy recession speed is 210 km/s. Use H0=70 km/s/Mpc and the nearby linear Hubble law to find distance.

answer: 3 Mpc (2 points).

Numerical target 3; absolute tolerance 0.006. Accepted units: , Mpc.

16. (2 points) A normalized binary exercise has separation 2 AU and period 2 years. Use total mass in solar masses=a^3/P^2. Find total mass.

answer: 2 solar masses (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , solar masses.

17. (2 points) NASA identifies M82 X-2 as an ultraluminous pulsar. What does a coherent rotation-linked pulse train imply about the compact object, and why is luminosity alone insufficient?

answer: The pulsing compact object is a rotating neutron star. High X-ray luminosity by itself does not uniquely establish that a source is a black hole. (2 points).

18. (2 points) SN 2014J is a Type Ia supernova in M82. Explain why standardized Type Ia luminosities can constrain distance and why dust must be considered.

answer: Compare extinction-corrected apparent brightness with calibrated intrinsic luminosity; dust changes observed brightness and color and can bias inferred distance. (2 points).

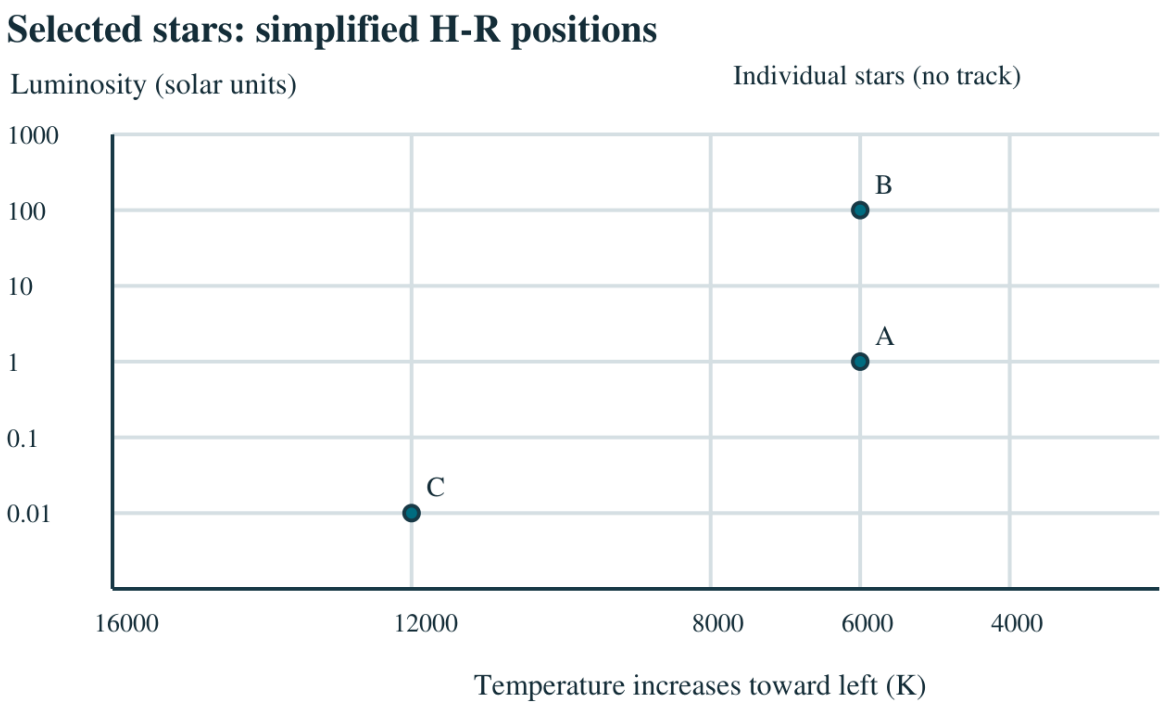
19. (2 points) GW170817 was observed in gravitational waves and electromagnetic radiation. Explain the value of this multi-messenger evidence for a neutron-star merger.

answer: The signals independently constrain binary inspiral and luminous ejecta/afterglow, linking compact-object dynamics to an electromagnetic counterpart rather than relying on one signal alone. (2 points).

20. (2 points) Synthetic starburst-galaxy case 3: ultraviolet emission and infrared dust emission both increase. Explain why using only visible light can miss part of the recent star formation.

answer: Dust can absorb shorter-wavelength stellar radiation and reradiate energy in the infrared. Multi-wavelength measurements help constrain obscured young populations; separate other heating sources. (2 points).

21. (2 points) Use Figure F1. Use L proportional to R^2 T^4 to find radius B / radius A.



F1. Hertzsprung-Russell diagram (original synthetic teaching figure)

answer: 10 (2 points).

Numerical target 10; absolute tolerance 0.03. Accepted units: .

22. (2 points) Use Figure F1. Which labeled object best fits a hot, low-luminosity compact remnant position? Explain.

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

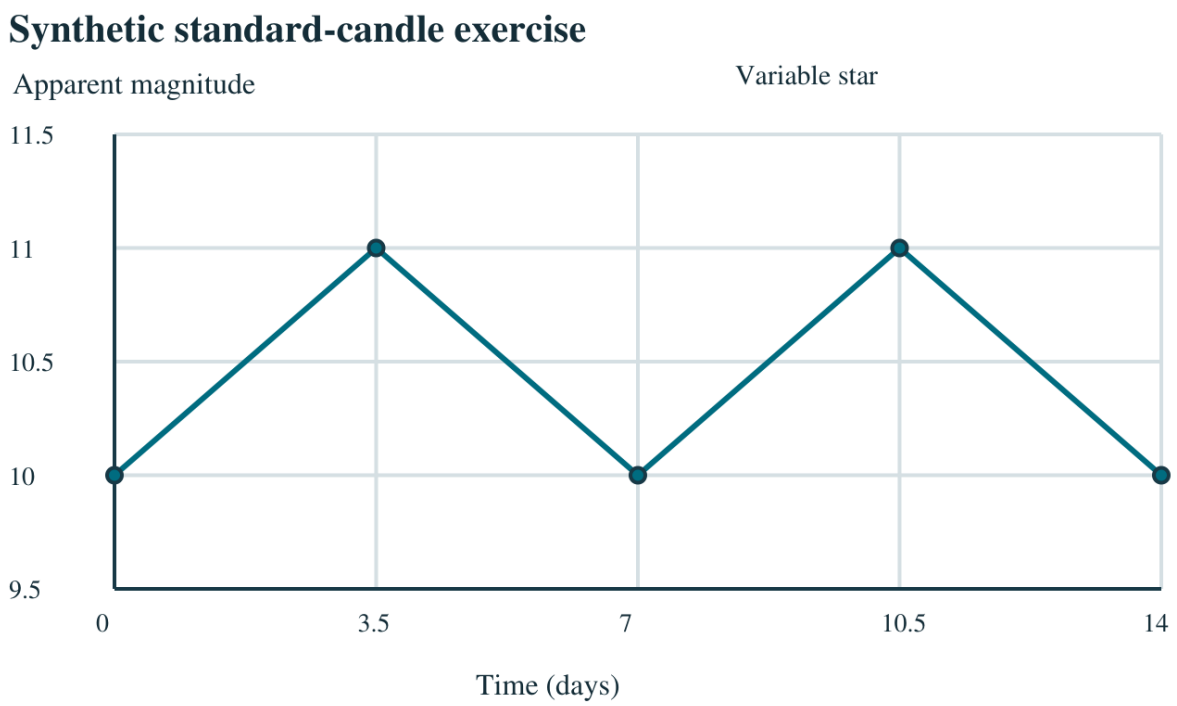
answer: C; high temperature but very low luminosity implies a small radiating surface, consistent with a white-dwarf-like position. (2 points).

23. (2 points) Use Figure F1. Why should the displayed positions not be read as motion of one star through space?

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

answer: The points represent different objects in temperature-luminosity space; their placement is not a spatial path or a demonstrated evolution sequence. (2 points).

24. (2 points) Use Figure F2. Read the repeat period.



F2. Pulsating-variable photometry (original synthetic teaching figure)

answer: 7 days (2 points).

Numerical target 7; absolute tolerance 0.021. Accepted units: , days.

25. (2 points) Use Figure F2. For this teaching estimate use the arithmetic midpoint of the plotted magnitude extrema as m. Take M=0.5, negligible extinction, and m-M=5log10(d/10pc). Find distance.

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1000 pc (2 points).

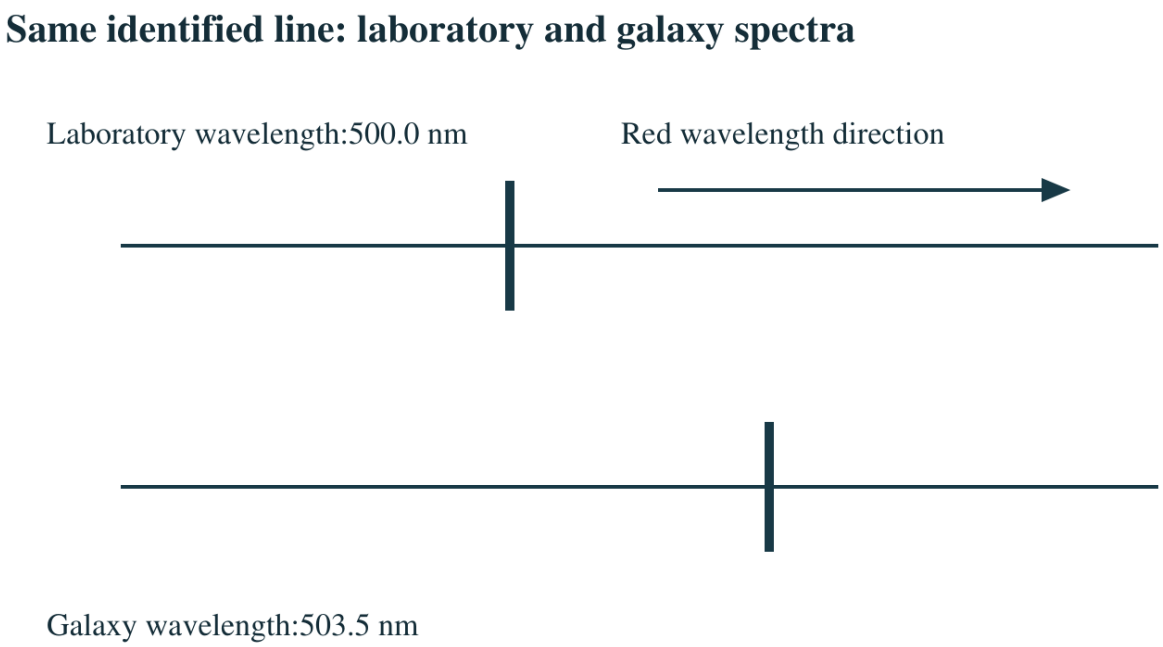
Numerical target 1000; absolute tolerance 3. Accepted units: , pc.

26. (2 points) Use Figure F2. At plotted magnitude10 versus11, which state is brighter and why?

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

answer: Magnitude10 is brighter; astronomical magnitude decreases as received flux increases. (2 points).

27. (2 points) Use Figure F3. Calculate z=(observed-rest)/rest.



F3. Spectral-line displacement (original synthetic teaching figure)

answer: 0.007 (2 points).

Numerical target 0.007; absolute tolerance 2.1e-05. Accepted units: .

28. (2 points) Use Figure F3. Use the low-redshift relation v=cz, with c=300000 km/s, to estimate recession speed.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

answer: 2100 km/s (2 points).

Numerical target 2100; absolute tolerance 6.3. Accepted units: , km/s.

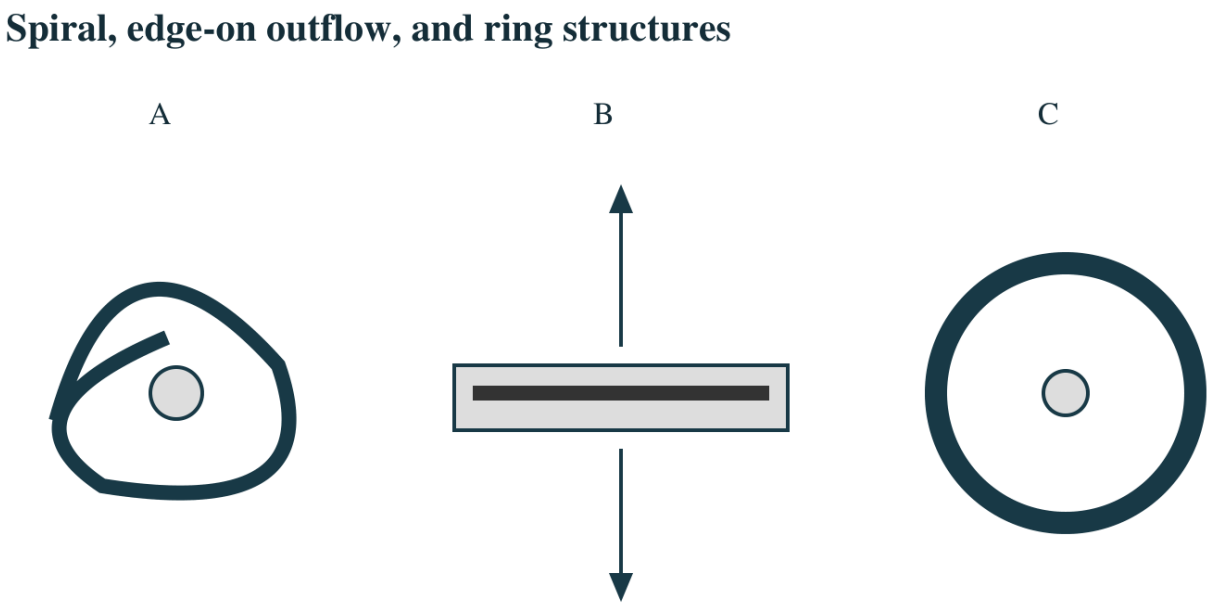
29. (2 points) Use Figure F3. Use H0=70 km/s/Mpc and the nearby linear Hubble law to estimate distance.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

answer: 30 Mpc (2 points).

Numerical target 30; absolute tolerance 0.09. Accepted units: , Mpc.

30. (2 points) Use Figure F4. Which panel visibly shows a ring morphology, relevant when studying the listed Cartwheel Galaxy?



F4. Galaxy morphology comparison (original synthetic teaching figure)

answer: C. (2 points).

31. (2 points) Use Figure F4. Which panel illustrates an outflow perpendicular to an edge-on disk, a geometry relevant to starburst-galaxy studies?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

answer: B. (2 points).

32. (2 points) Use Figure F4. Why cannot these three morphology sketches alone establish a universal galaxy age sequence?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

answer: Morphology depends on structure, interactions, viewing orientation and history; the sketches provide no universal time ordering or complete population evidence. (2 points).